Measures of Center & Spread

ANSWER KEY

Section 1: Mean, Median, and Skewness

1 In a strongly right-skewed distribution of household incomes, which relationship between the mean and median is typically expected?

a. **The mean is greater than the median ✓**

b. The mean is less than the median

c. The mean equals the median

d. The relationship cannot be determined without sample size

In right-skewed distributions, unusually large values pull the mean toward the right tail, making the mean greater than the median.

2 Which summary statistic is considered resistant to extreme outliers?

a. **Median and Interquartile Range (IQR) ✓**

b. Mean and Standard Deviation

c. Range and Mean

d. Standard Deviation and Variance

The median and IQR are based on order statistics/ranks and are resistant to extreme outliers, unlike the mean and standard deviation which incorporate every numeric value.

3 A dataset has values: $3, 5, 7, 8, 9, 11, 15$. If the maximum value 15 is replaced with 50, which statistic will NOT change?

a. **Median ✓**

b. Mean

c. Range

d. Standard Deviation

The median is the 4th value (8). Changing the extreme value from 15 to 50 leaves the order and the middle value (8) unchanged.

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Section 2: The 1.5 x IQR Rule for Outliers

4 A dataset has $Q_1 = 24$ and $Q_3 = 40$. According to the $1.5 \times \text{IQR}$ rule, what are the lower and upper outlier boundaries?

a. **Lower:** 0**; Upper:** 64 ✓

b. Lower: 8; Upper: 56

c. Lower: 16; Upper: 48

d. Lower: 12; Upper: 60

$\text{IQR} = Q_3 - Q_1 = 40 - 24 = 16$. $1.5 \times \text{IQR} = 24$. Lower boundary: $Q_1 - 24 = 24 - 24 = 0$. Upper boundary: $Q_3 + 24 = 40 + 24 = 64$.

5 If a distribution has five-number summary: $\text{Min} = 10$, $Q_1 = 20$, $\text{Med} = 28$, $Q_3 = 35$, $\text{Max} = 62$, which value is an outlier?

a. 62 ✓

b. 10

c. Both 10 and 62

d. Neither

$\text{IQR} = 35 - 20 = 15$. Upper threshold: $35 + 1.5(15) = 35 + 22.5 = 57.5$. Lower threshold: $20 - 22.5 = -2.5$. Since $62 > 57.5$, 62 is an upper outlier. $10 > -2.5$, so 10 is not an outlier.

6 What percentage of observations in any distribution lies strictly between Q_1 and Q_3?

a. **Approximately** 50% ✓

b. 25%

c. 75%

d. 68%

By definition, Q_1 is at the 25th percentile and Q_3 is at the 75th percentile, so exactly the middle 50% of data lies within the interquartile range.

Section 3: Standard Deviation and Variance

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7 What does sample standard deviation s measure in a dataset?

a. **The typical distance of data points from the sample mean** ✓

b. The distance between the maximum and minimum observations

c. The difference between the mean and median

d. The percentage of values that are negative

Standard deviation measures the typical (or average) distance of observations from the arithmetic mean.

8 If every observation in a sample has the exact same value of 14, what is the sample standard deviation?

a. 0 ✓

b. 14

c. 1

d. Undefined

When all observations are identical, there is zero variability, so each deviation from the mean ($14 - 14 = 0$) is zero, resulting in $s = 0$.

9 Why is $(n - 1)$ used in the denominator of the sample variance $s^2 = \frac{\sum(x_i - \bar{x})^2}{n - 1}$ instead of n?

a. **To make s^2 an unbiased estimator of the population variance σ^2** ✓

b. To ensure the standard deviation is always an integer

c. Because one observation is always assumed to be an outlier

d. To adjust for potential measurement errors in the sample

Dividing by $n - 1$ provides degrees of freedom correction so that the expected value of s^2 equals σ^2, making it an unbiased estimator.

Section 4: Comparing Distributions with Boxplots

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10 A boxplot shows a long right whisker, a right-half box wider than the left-half box, and no outliers. What is the shape of the distribution?

a. **Skewed to the right ✓**

b. Skewed to the left

c. Roughly symmetric

d. Uniform

An elongated right whisker and a wider upper half of the box indicate that values in the upper 50% are spread over a wider interval, characteristic of a right-skewed distribution.

11 When comparing two test score distributions, School A has $\text{median} = 78, \text{IQR} = 12$ and School B has $\text{median} = 85, \text{IQR} = 6$. Which statement is justified?

a. **School B typically scored higher and exhibited less variability in the middle 50% of scores ✓**

b. School A had a higher maximum score than School B

c. School A had higher consistency in student scores

d. School B had more outliers than School A

School B's higher median ($85 > 78$) indicates higher central tendency, and its smaller IQR ($6 < 12$) shows less spread and higher consistency among the middle 50%.

12 Which display allows you to recover the original raw data values?

a. **Stem-and-leaf plot ✓**

b. Boxplot

c. Histogram

d. Cumulative relative frequency plot

A stem-and-leaf plot retains all individual data values in its stems and leaves, whereas boxplots and histograms bin or summarize data into aggregates.

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